ECE 6580 — Module Project 2
Option A: Eddy Currents via FEMM

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Abstract

The objective of this report is to model an eddy-current displacement transducer by using the finite element solver FEMM on an axisymmetric domain. A copper coil is driven with AC current at 10 khz above an aluminum plate the lift-off gap is swept from 0 mm to 10 mm, and the coil impedance response is used to quantify how eddy-current effects change with distance. A coupling coefficient $k(z)$ is derived from the inductance change and compared with the module's analytical distance dependence (Eq. (72)) to evaluate agreement and discuss sources of derivation.

1 Introduction

Eddy-current sensors are used as a standard approach for *non-contact* displacement measurement. In manufacturing and robotics, eddy-current sensors can be use to measure gap and runout. Moreover, in vibration monitoring, they are used to track shaft motion. Eddy-current sensors are used in nondestructive evaluation can detect cracks and changes in conductivity. Their appeal is that the sensing mechanism is electromagnetic rather than mechanical, so the sensor can operate through thin coatings and does not require physical contact with the target.

The physical mechanism is straightforward and a time-varying current in the coil produces a time-varying magnetic field. When a conducting target, is nearby, Faraday's law implies an induced electric field that drives closed-loop currents in the conductor (eddy currents). Those induced currents dissipate real power through Joule heating and generate an opposing magnetic field (Lenz's law). Both effects are observable at the coil terminals as a change in complex impedance:

$$Z(z) = \frac{V}{I} = R(z) + jX(z)$$

When the target is close, eddy currents are strong and the reflected loss into the coil increases, so $R(z)$ is larger. At the same time, the opposing field reduces net flux linkage and the coil inductance is suppressed. When the target is far away, eddy currents weaken and the coil approaches its "free-space" behavior.

Analytical solutions exist for idealized geometries and are useful for intuition. A classic reference is the probe-coil treatment by Dodd and Deeds (1968), which derives closed-form expressions for eddy-current probe problems. However, real geometries include finite coil thickness, finite plate thickness, and boundary truncation effects. For this reason, simulation tools are widely used for quantitative predictions; this project uses FEMM (Meeker, 2015).

The objectives of this report are as follows:

- Option A-a: Demonstrate via simulation that eddy-current effects change with coil–target distance (lift-off).
- Option A-b: Compute a coupling coefficient from the simulation, plot Eq. (72) using simulation parameters, and compare theory vs simulation.

2 Theory and Metrics

2.1 Impedance Response and Inductance Extraction

The coil impedance is computed from the circuit quantities returned by FEMM and is given as follows:

$$Z(z) = \frac{V}{I} = R(z) + jX(z). \tag{1}$$

The resistive part $R(z)$ denotes a proxy for eddy-current loss because it represents real power drawn from the source for a given current (beyond the ideal lossless inductive behavior) and the reactive part $X(z)$ corresponds to energy storage in the magnetic field. Also, the inductance is obtained from $X(z) = \omega L(z)$:

$$L(z) = \frac{X(z)}{2\pi f}. \tag{2}$$

In a coil-over-conductor configuration, $L(z)$ typically increases with lift-off because the opposing field generated by eddy currents weakens with distance.

2.2 Skin Depth and Why Frequency Matters

The spatial distribution of induced currents depends strongly on frequency through the *skin-depth*. In a good conductor with permeability μ and conductivity σ, a common estimate is given by:

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$$

For aluminum ($\mu \approx \mu_0$, $\sigma \approx 3.5 \times 10^7$ S/m) at 10 kHz, and this gives $\delta \approx 0.85$ mm. Since the plate thickness in this project is 5 mm, the thickness is several skin depths at 10 Khz. This means that most induced current density is concentrated near the surface closest to the coil, and the impedance response is dominated by near-surface diffusion rather than currents that are distributed through the full plate thickness. This also helps to explain why sensor response can be very sensitive to small lift-off changes near the surface.

2.3 Coupling Coefficient from Simulation

To obtain a normalized coupling-like quantity from simulation, we can use the inductance change with lift-off. Let L_∞ denote the inductance at sufficiently large separation (approximated here by the largest gap in the sweep). Define

$$k^2(z) \equiv \frac{L_\infty - L(z)}{L_\infty - L(0)}, \qquad k(z) = \sqrt{k^2(z)}. \tag{3}$$

This normalization enforces $k^2(0) = 1$ and $k^2(z) \to 0$ as z increases. Operationally, it treats the presence of the conductive plate as a "shorted secondary" that draws energy from the primary in proportion to coupling.

2.4 Module Eq. (72) and Normalized Comparison

The course notes of ECE 6580 (Course Staff, 2026) propose the following distance dependence (Eq. (72)) as a simplified model for coupling-like behavior:

$$M^2(z) = e^{-z}\,\frac{L_1\,\mu\,S\,b^2}{2\,(z^2+b^2)^{3/2}}. \tag{4}$$

Here, L_1 is the nominal coil inductance, μ is permeability, b is a characteristic coil radius (taken as the mean coil radius), and $S = \pi b^2$ is the corresponding area. Because Eq. (4) is used primarily for *shape* vs distance in the module discussion,

this report compares *normalized* curves (each divided by its value at $z = 0$) to assess similarity of distance dependence, rather than absolute magnitude.

3 Simulation Setup (FEMM)

FEMM (Meeker, 2015) solves low-frequency electromagnetic problems on 2D planar and axisymmetric domains and FEMM scripting utilizes a Lua 4 variant with complex numbers (ComplexLua) (Meeker, 2024), which is well-suited to automate parametrer sweeps and postprocessing of complex impedance.

3.1 Geometry and Materials

The geometry follows the "coil over conducting plate" configuration:

- **Coil**: Copper, inner radius 4.5 mm, outer radius 6.5 mm, height 2 mm (mean radius $b = 5.5$ mm).
- **Plate**: Aluminum 6061-T6, radius 50 mm, thickness 5 mm (top surface at $z = 0$).
- **Air Region**: Radius 80 mm, vertical extent approximately -30 mm to 50 mm.

The coil is excited by an AC current source at 10 kHz. Lift-off gap z is defined as the distance from the plate surface to the bottom of the coil and is swept from 0 mm to 10 mm in 0.5 mm steps. The excitation current is set to 1 A with 100 turns so that induced effects are visible and impedance extraction is straightforward.

3.2 Boundary Conditions and Meshing

A Dirichlet boundary $A = 0$ is applied on the outer air box to approximate an open region. The air box is chosen significantly larger than the coil so that boundary truncation has minimal influence on the near-field coupling. The mesh is refined near the coil and plate where field gradients and induced currents are strongest, and coarser elsewhere to reduce solve time while retaining accuracy in the region of interest.

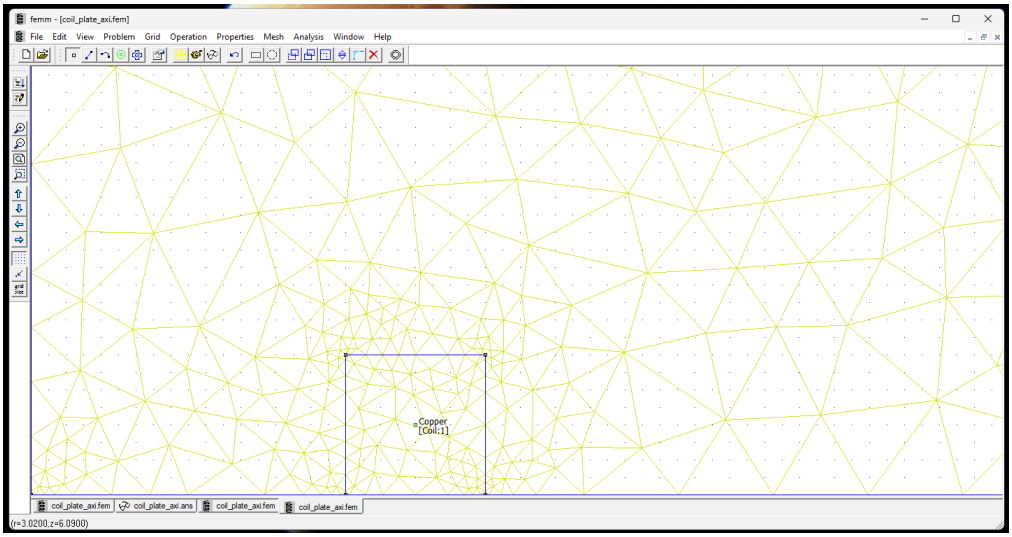


Figure 1: FEMM mesh near the coil and plate. (Use `Mesh Geometry_Lua.png`.)

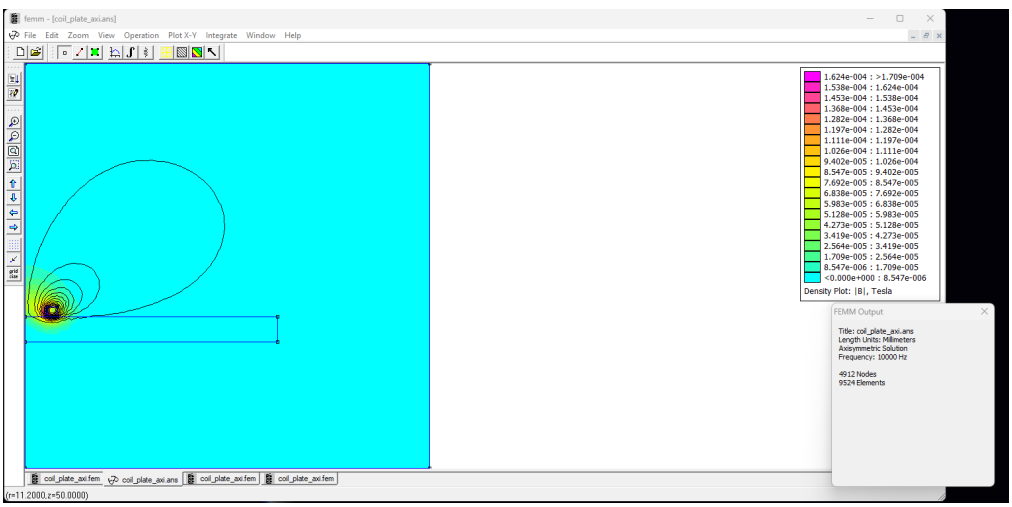
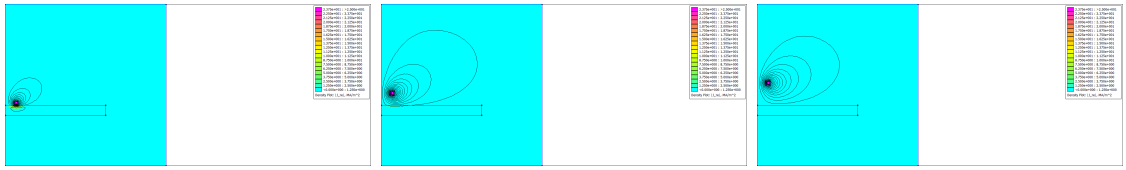


Figure 2: Example FEMM field visualization (magnetic flux density $|B|$). (Use `Density Contour Plot_Lua .png`.)



(a) $z = 0$ mm (b) $z = 5$ mm (c) $z = 10$ mm

Figure 3: Eddy-current density ($\mathrm{Re}(J)$) in the aluminum plate decreases with lift-off distance. These plots directly satisfy the "show how eddy currents change with distance" requirement.

3.3 Postprocessing and Data Reduction

For each gap value, the FEMM solution provides circuit values for the named coil circuit, from which $Z = V/I$ is computed. The sweep data are stored in `coil_plate_sweep.csv` with columns `gap_mm`, `R_ohm`, `X_ohm`, and `L_H`. Additional derived quantities (e.g., $k(z)$) are computed from $L(z)$ using the normalization described above. Plotting $R(z)$ provides a direct indicator of eddy-current loss, while plotting $L(z)$ provides a compact summary of the coupling reduction with lift-off.

4 Results: Eddy Currents Change with Distance (Option A-a)

Eddy-current effects are observed visually in the field plot (Fig. 2), directly in the plate current-density maps (Fig. 3), and quantitatively through the coil impedance response. As lift-off increases, the induced current density in the conductor decreases, reducing real power dissipation. This appears as a reduction in the resistive component $R(z)$ of the coil impedance.

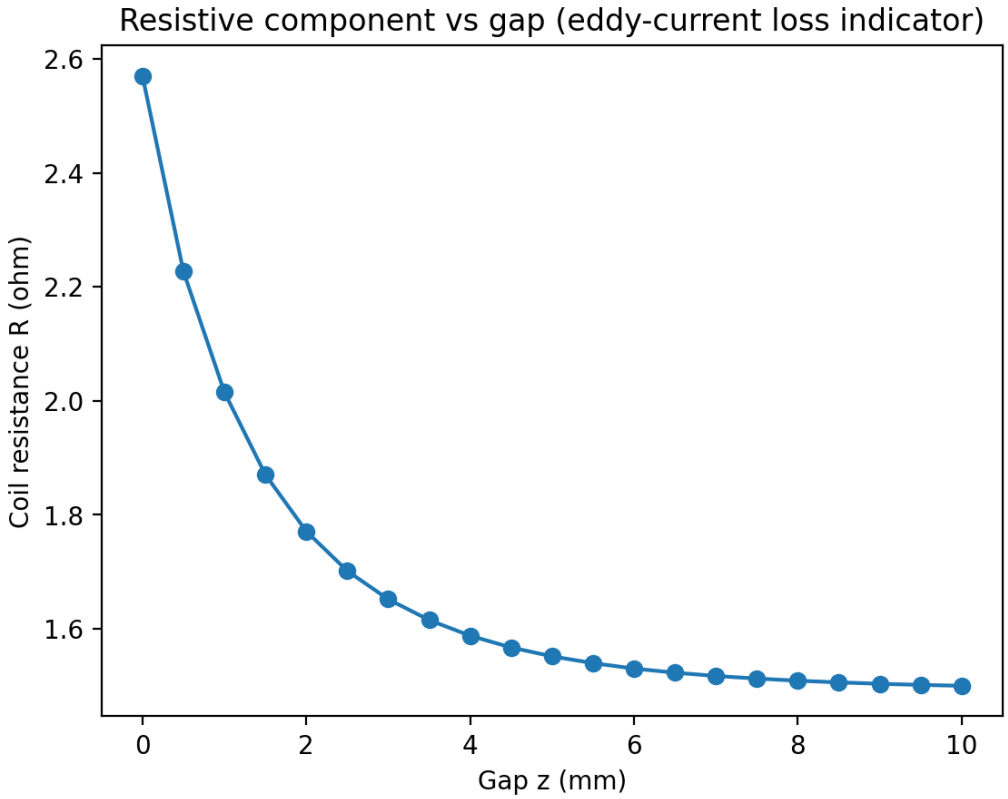


Figure 4: Resistive component of coil impedance $R(z)$ vs lift-off gap. Larger R indicates stronger eddy-current loss when the plate is close.

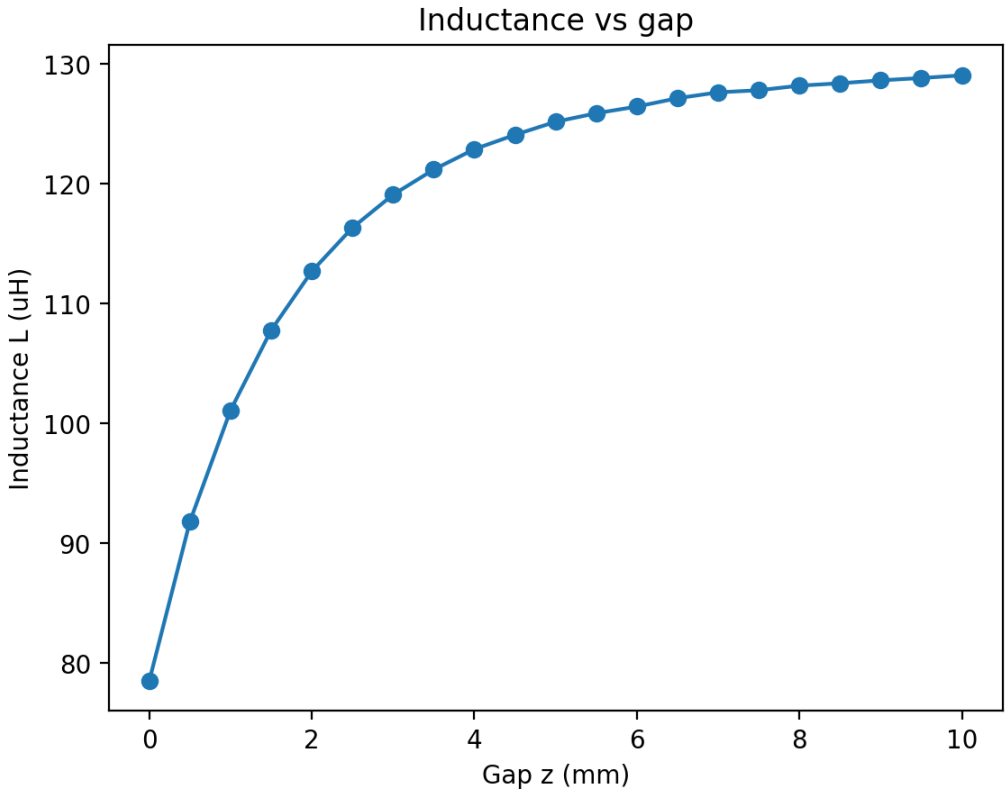


Figure 5: Inductance $L(z)$ vs lift-off gap. Inductance increases toward a free-space value as coupling to the conductive plate weakens with distance.

4.1 Quantitative Trends and Sensor Sensitivity

From the sweep, the coil resistance decreased from $R(0) \approx 2.570\,\Omega$ at $z = 0\,\text{mm}$ to $R(10) \approx 1.499\,\Omega$ at $z = 10\,\text{mm}$, consistent with reduced eddy-current power dissipation as coupling weakens. The inductance increased from $L(0) \approx 78.5\,\mu\text{H}$ to $L(10) \approx 129.0\,\mu\text{H}$, approaching the free-space value as the plate becomes electromagnetically "far." The total inductance change across the sweep is approximately $50.5\,\mu\text{H}$.

A key practical point for displacement sensing is *sensitivity*. Using a simple numerical derivative, the initial slopes are approximately

$$\left.\frac{dL}{dz}\right|_{z\approx 0} \approx 26.6\,\mu\text{H/mm}, \qquad \left.\frac{dR}{dz}\right|_{z\approx 0} \approx -0.69\,\Omega/\text{mm}.$$

These values emphasize that the strongest sensitivity occurs at small lift-off, which is typical of eddy-current probes: most useful measurement range is near the surface where induced currents are strongest. As z increases, both $R(z)$ and $L(z)$ curves gradually saturate, implying reduced incremental sensitivity at larger distances. In practice, sensors are often calibrated over a limited range (e.g., a few millimeters) to achieve a more linear mapping from impedance to displacement.

A subset of the sweep data is shown in Table 1 (full data in `coil_plate_sweep.csv`). The trend matches the eddy-current interpretation: $R(z)$ decreases with increasing lift-off as induced losses diminish, while $L(z)$ increases because the opposing field from eddy currents becomes weaker.

Table 1: Representative Sweep Results (subset of `coil_plate_sweep.csv`).

Gap z (mm)	R (Ω)	X (Ω)	L (μH)
0.0	2.5699	4.9336	78.5
1.0	2.0147	6.3520	101.1
2.0	1.7705	7.0805	112.7
3.5	1.6147	7.6140	121.2
4.5	1.5666	7.7963	124.1
5.5	1.5391	7.9094	125.9
6.5	1.5223	7.9882	127.1
8.0	1.5083	8.0541	128.2
9.0	1.5028	8.0819	128.6
10.0	1.4993	8.1081	129.0

5 Results: Coupling Coefficient and Eq. (72) Comparison (Option A-b)

Using the inductance-based normalization,

$$k^2(z) = \frac{L_\infty - L(z)}{L_\infty - L(0)},$$

we compute $k(z) = \sqrt{k^2(z)}$ across the lift-off sweep. The coupling coefficient decreases rapidly near the surface and gradually approaches zero at larger lift-off, consistent with the qualitative eddy-current picture: as the target moves away, induced currents weaken and the coil behaves more like a free-space inductor.

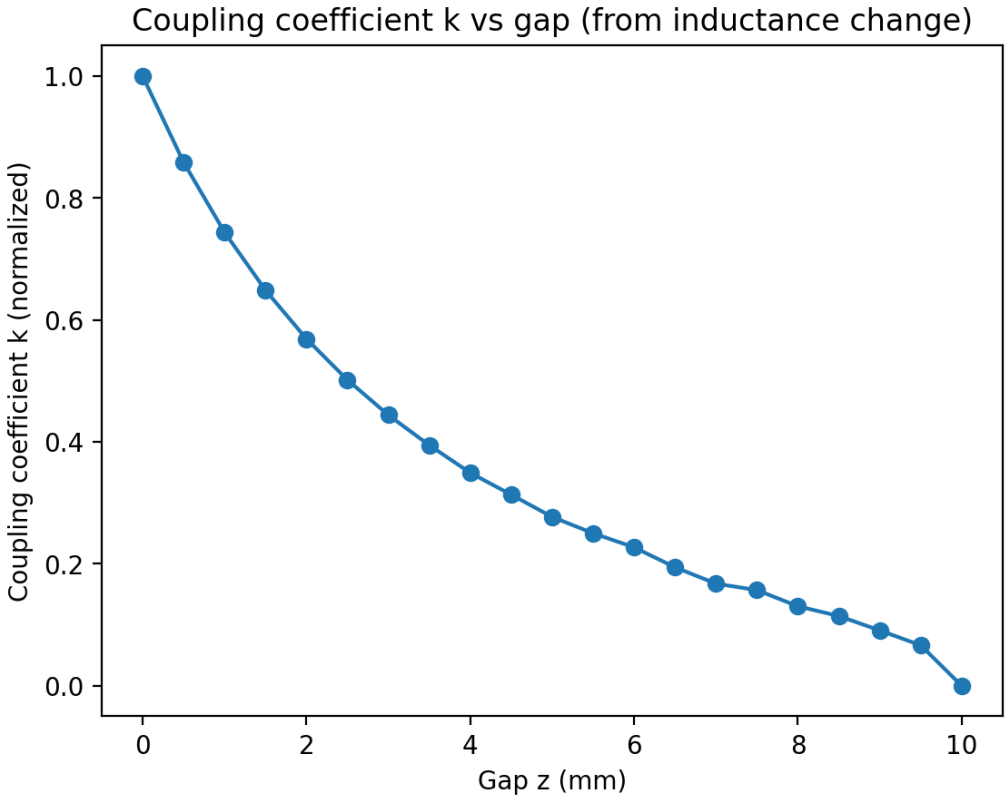


Figure 6: Coupling coefficient $k(z)$ computed from inductance change, normalized so that $k(0) = 1$.

To compare with the module derivation (Course Staff, 2026), Eq. (4) is evaluated using simulation parameters: $b = 5.5\,\mathrm{mm}$, $S = \pi b^2$, $\mu \approx \mu_0$ for aluminum, and $L_1 \approx L_\infty$. Because Eq. (4) is used as a simplified distance-dependence model, both the simulation-derived curve and the Eq. (72) curve are normalized by the $z = 0$ value before comparison.

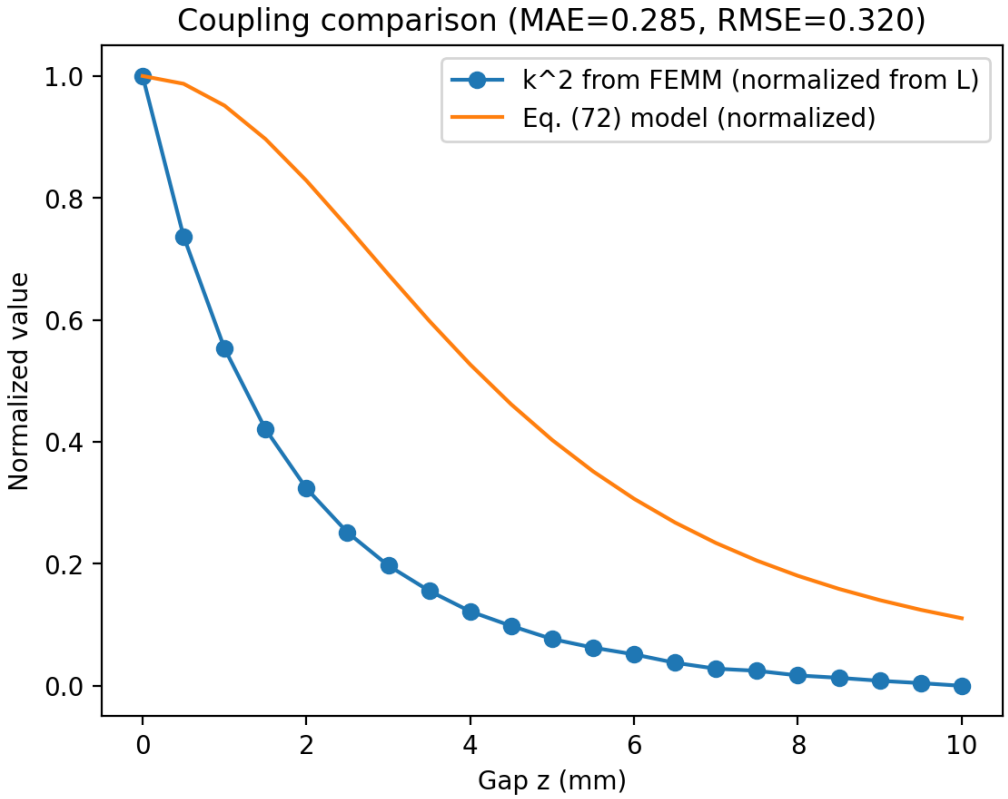


Figure 7: Normalized simulation-derived coupling (from $L(z)$) compared to normalized Eq. (72) curve.

5.1 Discussion: Agreement and Deviations

The theoretical curve and the simulation-derived curve share the expected qualitative behavior: strong coupling near the surface and decay with increasing lift-off. Quantitative differences are expected because Eq. (72) is derived under simplifying assumptions. Key sources of deviation include:

- **Finite coil geometry:** Eq. (72) uses a single characteristic radius b (thin-loop approximation). The simulated coil has finite radial thickness and height, altering field distribution and the distance dependence of coupling.
- **Finite plate thickness and diffusion:** Induced currents are not uniform through the plate; they decay roughly over the skin depth. FEMM captures

current diffusion and the dependence on σ and f, while Eq. (72) collapses these details into a simple distance dependence.

- **Heuristic exponential factor:** The module notes incorporate a simple exponential decay $e^{[-0.-0.5-1.-1.5-2.-2.5-3.-3.5-4.-4.5-5.-5.5-6.-6.5-7.-7.5-8.-8.5-9.-9.5-10.]}$ for $k^2(z)$. In reality the decay is governed by the full field solution, and may not be strictly exponential across the full range.

- **Boundary truncation and numerical factors:** FEMM uses a finite computational domain and a finite mesh. Provided the air region is sufficiently large and the mesh is refined near the coil and plate (Fig. 1), these effects should be small but can still contribute to mismatch at larger distances.

5.2 Practical Interpretation for a Displacement Sensor

From a transducer design standpoint, two behaviors are most important:

1. Monotonicity
2. A usable sensitivity range

The simulation produces monotonic $R(z)$ and $L(z)$ curves, indicating an unambiguous mapping from lift-off to impedance in the modeled range. The strongest sensitivity occurs close to the target, so a practical sensor would typically be calibrated and operated in a limited lift-off interval where the slope is relatively consistent. The diminishing slope at larger gaps indicates the measurement becomes less sensitive and more susceptible to noise, drift, and modeling uncertainty. This motivates the common engineering practice of selecting coil dimensions and operating frequency so that the intended measurement range aligns with the high-sensitivity portion of the curve.

6 Additional Analysis: Skin Depth and Modeling Considerations

A useful physical scale in eddy-current problems is the *skin depth* δ, the characteristic distance over which induced currents decay inside a conductor. For a good conductor

with conductivity σ and permeability μ, the skin depth at angular frequency $\omega = 2\pi f$ is approximately

$$\delta \approx \sqrt{\frac{2}{\omega\mu\sigma}}. \tag{5}$$

For aluminum, $\mu \approx \mu_0$ and σ is on the order of 3×10^7 S/m, which at $f = 10\,\text{kHz}$ yields $\delta \sim 0.8\,\text{mm}$. Because the simulated plate thickness is 5 mm, the plate is several skin depths thick at the operating frequency, meaning most induced current is concentrated near the surface facing the coil. This helps explain why the coil resistance is most sensitive to lift-off at small gaps: the strongest coupling occurs to the near-surface eddy currents that directly oppose the coil field.

From a numerical perspective, the mesh must be fine enough near the coil and the plate surface to resolve these gradients. The mesh shown in Fig. 1 is intentionally denser around the coil and the plate interface. While a full mesh-convergence study is beyond the scope of this module, a practical check is that the extracted circuit quantities (R, X, and L) vary smoothly with z and do not exhibit nonphysical oscillations with step size. The monotonic trends in Figs. 4–5 and the smooth coupling decay in Fig. 6 support that the discretization is capturing the dominant physics.

7 Automation and Data Extraction Workflow

To obtain the response curves efficiently, the FEMM model was scripted to sweep the lift-off gap z over the specified range. For each gap value, the workflow is:

1. Build or Reposition the Coil Geometry
2. Generate the Mesh
3. Solve the AC Magnetics Problem
4. Load the post-processing solution the circuit properties for the excitation circuit named `Coil`.
5. Query the circuit properties for the excitation circuit named `Coil`.

FEMM provides the complex current I and voltage V for the circuit, enabling the impedance calculation $Z = V/I$. The exported CSV file (`coil_plate_sweep.csv`) stores $R = \Re\{Z\}$, $X = \Im\{Z\}$, and $L = X/(2\pi f)$ at each gap. This automated sweep ensures that the plotted trends are generated from a consistent model configuration and reduces the chance of transcription error compared to manual measurement for each gap value.

8 Conclusion

This project modeled a coil-over-plate eddy-current transducer using FEMM on an axisymmetric domain at 10 kHz. The simulation demonstrated that eddy-current effects decrease with lift-off: the resistive component $R(z)$ decreases with distance (reduced eddy-current loss), and the inductance $L(z)$ increases toward a free-space value. A coupling coefficient $k(z)$ derived from inductance change shows monotonic decay with lift-off and was compared to the module's Eq. (72) curve using simulation-derived parameters. Qualitative agreement is observed, while quantitative differences are attributed to simplifying assumptions in the derivation and full-physics effects captured by FEMM.

References

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